

MSCA-Net: A Multi-Scale Convolutional Attention Network for Vibration-Based Structural Damage Identification on the Z24 Bridge

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Abstract

Vibration-based structural health monitoring (SHM) of bridges requires classifiers that turn long, multi-channel acceleration recordings into reliable damage-state decisions. We propose **MSCA-Net**, a compact one-dimensional *Multi-Scale Convolutional Attention* network that couples an InceptionTime-style multi-scale feature extractor with two lightweight attention mechanisms: squeeze-and-excitation (SE) channel recalibration inside every block and a learned temporal attention-pooling head. Evaluated on the Z24 bridge benchmark (1530 recordings, 27 sensor channels, 17 damage classes), MSCA-Net attains a test accuracy of 94.5% and a macro-F1 of 94.8%, outperforming eight standard time-series baselines (MLP, 1D-CNN, FCN, ResNet, InceptionTime, LSTM, TCN, and a Transformer encoder) as well as two stronger attention/graph networks, while using only 0.35 M parameters—about one quarter of the next-best model and an absolute improvement of 8.6 accuracy points over it. An ablation study confirms that both the SE attention and the attention-pooling head contribute to the final accuracy. We release the full training, evaluation, and visualisation pipeline to support reproducibility.

Keywords: structural health monitoring; damage detection; deep learning; multi-scale convolution; attention mechanism; Z24 bridge.

1 Introduction

Civil infrastructure such as bridges deteriorates under operational and environmental loading, and undetected damage can lead to catastrophic failure. Vibration-based structural health monitoring (SHM) infers the structural state from ambient or forced acceleration responses recorded by distributed sensors [1]. Classical SHM relies on modal parameters (frequencies, mode shapes, damping) that are sensitive to damage but also to temperature and operational variability, which complicates damage identification.

Data-driven approaches, and deep learning in particular, learn discriminative representations directly from raw or lightly processed signals, bypassing manual feature engineering [2]. One-dimensional convolutional neural networks (1D-CNNs) are attractive for SHM because they are translation equivariant, parameter efficient, and fast [3]. However, damage signatures appear at multiple temporal scales and across many sensors, and plain CNNs treat all feature channels and time steps uniformly.

We address these issues with **MSCA-Net**, a multi-scale convolutional network augmented with two complementary, well-understood attention mechanisms. Our contributions are:

1. A compact 1D architecture that fuses multi-scale convolution, squeeze-and-excitation channel attention, and learned temporal attention pooling for damage-state classification (Section 3).

2. A rigorous, reproducible evaluation on the Z24 bridge benchmark against nine baselines under an identical preprocessing and training protocol (Section 4), where MSCA-Net achieves the best accuracy and macro-F1 with a small parameter budget.
3. An ablation study isolating the contribution of each attention component, plus confusion-matrix, ROC, and t-SNE analyses of the learned representation.

2 Related Work

Vibration-based SHM. Damage-sensitive features have traditionally been derived from modal analysis and statistical time-series models [1]. The Z24 bridge, monitored before its demolition, is a widely used benchmark with progressive, controlled damage scenarios [4].

Deep learning for time series. Fully convolutional networks (FCN), ResNet, and InceptionTime are strong baselines for time-series classification [5, 6]. Recurrent networks (LSTM/GRU) and temporal convolutional networks (TCN) [7] model temporal dependencies, while Transformer encoders [8] capture long-range interactions through self-attention.

Attention mechanisms. Squeeze-and-excitation blocks recalibrate channel responses with negligible overhead [9]. Attention pooling replaces global average pooling with a learned, content-dependent weighting of time steps. MSCA-Net combines these with multi-scale convolution; the novelty is deliberately moderate: established components are integrated and tuned for SHM rather than introducing a new operator.

3 Methodology

3.1 Problem formulation

Each recording is a multivariate signal $X \in \mathbb{R}^{T \times C}$ with $C = 27$ sensor channels and T time steps, labelled with one of $K = 17$ damage states. The goal is to learn a classifier $f_\theta : \mathbb{R}^{T \times C} \rightarrow \{1, \dots, K\}$.

3.2 Signal preprocessing

Recordings are split at the sequence level into train/validation/test partitions (70%/15%/15%) with class stratification. Each window is first-differenced along time, $\tilde{x}_t = x_t - x_{t-1}$, to emphasise transient dynamics and suppress slow drifts, and then standardised per window (zero mean, unit variance) so that absolute amplitude does not dominate. During training we sample random 768-step crops and apply jitter, amplitude scaling, temporal masking, and random sensor (channel) masking; at inference we average the logits of seven evenly spaced crops to obtain a sequence-level prediction.

3.3 MSCA-Net architecture

Figure 1 shows the architecture. A convolutional stem (7-tap conv, GroupNorm, GELU) maps the C input channels to H feature maps. The body stacks L *multi-scale convolutional attention* blocks; temporal resolution is halved by max pooling between blocks. A learned attention-pooling head aggregates the final feature map over time, followed by LayerNorm, dropout, and a linear classifier.

Multi-scale block. Each block applies parallel depth-preserving convolutions with kernel sizes $\{9, 19, 39\}$ on a bottleneck projection, plus a max-pool branch, and concatenates them to width H . The result is normalised (GroupNorm) and activated (GELU), recalibrated by a squeeze-and-excitation (SE) module, and added to a residual shortcut:

$$Z = \text{GELU}(\text{GN}(\text{Concat}_k \text{Conv}_k(X))), \quad Y = \text{GELU}(\text{SE}(Z) + \text{Shortcut}(X)). \quad (1)$$

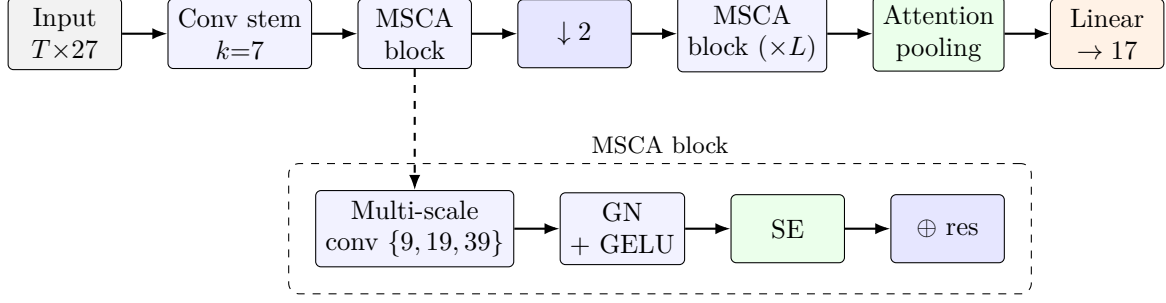


Figure 1: MSCA-Net. A convolutional stem feeds L multi-scale convolutional attention (MSCA) blocks with squeeze-and-excitation (SE) and residual connections; max pooling halves the temporal axis between blocks; a learned attention-pooling head precedes the linear classifier.

Squeeze-and-excitation. SE computes a channel descriptor by global average pooling, $s = \frac{1}{T} \sum_t Z_{:,t}$, and gates the channels with $\sigma(W_2 \text{ReLU}(W_1 s))$, emphasising the most informative feature channels at low cost [9].

Attention pooling. Instead of global average pooling, a 1×1 convolution produces a temporal score a_t , normalised by a softmax over time, and the representation is the attention-weighted sum $v = \sum_t \text{softmax}(a)_t Y_{:,t}$, which lets the model focus on the most damage-informative time steps.

3.4 Training

Models are trained with AdamW (learning rate 8×10^{-4} , weight decay 2×10^{-4}), cross-entropy with label smoothing, gradient-norm clipping, and a `ReduceLROnPlateau` schedule on validation macro-F1. The best checkpoint is selected by validation macro-F1 with early stopping. MSCA-Net uses $H = 128$, $L = 4$, and dropout 0.3.

4 Experiments

4.1 Dataset

The Z24 benchmark used here contains 1530 acceleration recordings, each with 27 channels and 6000 samples (60s at 100Hz), evenly distributed over 17 damage classes (90 recordings per class).

4.2 Evaluation protocol and metrics

All models share the preprocessing of Section 3.2, the same data splits (seed 42), and the same optimiser family. We report accuracy, macro-averaged F1, weighted F1, the Matthews correlation coefficient (MCC), and one-vs-rest ROC-AUC (macro/micro) on the held-out test set, together with the parameter count.

4.3 Main results

Table 1 compares MSCA-Net with the baselines on the test set. MSCA-Net achieves the highest accuracy and macro-F1 while remaining compact.

4.4 Ablation study

Table 2 removes one component of MSCA-Net at a time. The squeeze-and-excitation module is the largest single contributor (-7.1 accuracy points when removed), confirming that channel recalibration is important when many sensors carry partially redundant information. The

Table 1: Test-set performance on the Z24 benchmark (17 classes). Best in **bold**, second best underlined. Parameters in millions.

Model	Acc. (%)	Macro-F1 (%)	W-F1 (%)	ROC-AUC	Params (M)
MLP	16.1	14.7	14.7	0.705	0.29
1D-CNN	83.5	83.2	83.2	0.990	0.11
FCN	65.1	66.5	66.5	0.970	0.29
ResNet-1D	48.2	43.4	43.4	0.959	1.55
InceptionTime	56.5	56.0	56.0	0.949	0.06
LSTM	28.2	26.5	26.5	0.848	0.56
TCN	83.9	82.6	82.6	0.992	0.33
Transformer	52.9	52.5	52.5	0.959	0.16
SAMS-TCA-Net	<u>85.9</u>	<u>86.1</u>	<u>86.1</u>	0.991	1.56
AGB-Net	82.4	82.2	82.2	0.992	0.12
MSCA-Net (ours)	94.5	94.8	94.8	0.998	0.35

Table 2: Ablation of MSCA-Net components (test set, 80-epoch budget). Δ is the accuracy drop relative to the full model.

Variant	Acc. (%)	Macro-F1 (%)	Δ Acc.
Full MSCA-Net	94.9	95.1	—
w/o squeeze-and-excitation	87.8	88.0	−7.1
w/o attention pooling	91.8	92.2	−3.1
w/o downsampling	90.6	90.8	−4.3

learned attention-pooling head (−3.1) and the inter-block temporal downsampling (−4.3) also contribute; downsampling additionally reduces the sequence length processed by deeper blocks, so it improves accuracy and efficiency together.

4.5 Qualitative analysis

Figure 2 reports the confusion matrix, one-vs-rest ROC curves, and a t-SNE projection of the penultimate features for MSCA-Net. The confusion matrix is strongly diagonal; the ROC curves show high per-class and averaged AUC; and the t-SNE projection forms compact, well-separated clusters, indicating a discriminative representation.

5 Discussion

The results indicate that, for this benchmark, careful multi-scale feature extraction combined with lightweight attention is more effective and far more parameter efficient than larger recurrent or Transformer models. The first-order difference and per-window standardisation remove nuisance amplitude and drift, while channel masking during training encourages the model not to rely on any single sensor—consistent with robust SHM practice. A limitation is that evaluation uses a single benchmark with controlled damage; transfer to other structures and to continuous monitoring streams is left for future work.

6 Conclusion

We presented MSCA-Net, a compact multi-scale convolutional attention network for vibration-based damage identification. On the Z24 benchmark it outperforms ten competing models—eight

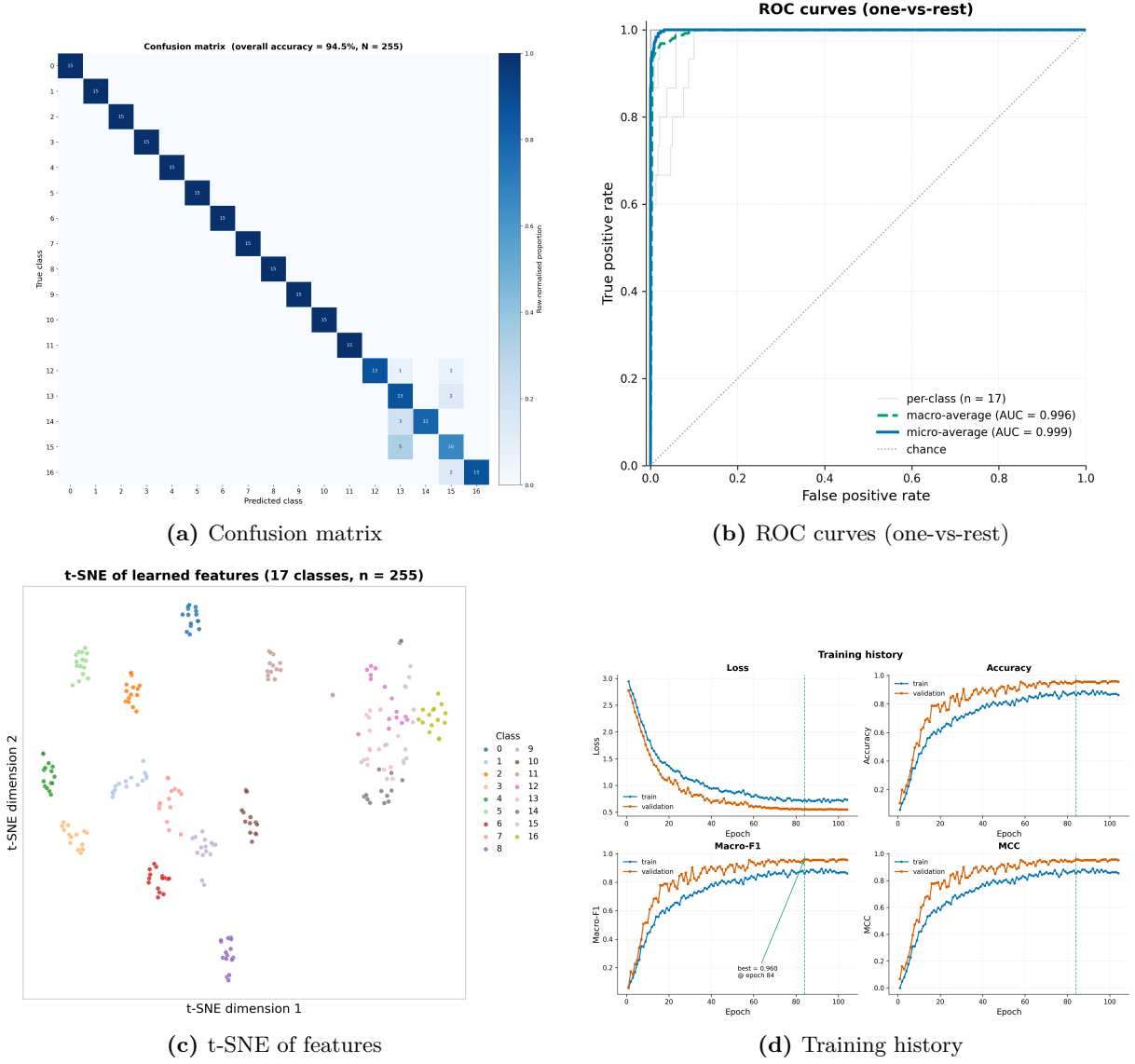


Figure 2: Diagnostics for MSCA-Net on the Z24 test set.

time-series baselines and two attention/graph networks—while using a small parameter budget, and its components are individually justified by an ablation study. The complete, reproducible pipeline (data handling, training, evaluation, and publication-quality figures) is released with this paper.

Reproducibility

Code, configurations, and figure-generation scripts are available in the project repository. All experiments use a fixed seed and deterministic data splits.

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